Acetone ⇒ Ketones

$CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

$-\overset{O}{\overset{\|}{C}}-$ Carbonyl group

* Physical Properties: →

- Color
- state
- Miscibility
- effect on litmus Paper

* Chemical Properties: →

1. Schiff's Reagent

→ 1 ml acetone
→ 1 ml schiff's reagent
(shaking) ⇓ Purple Color

2. Iodoform Test: →

→ 1 ml acetone
→ drops of Conc. NaOH (Pale-yellow Color)
→ 3 ml I_2 (Brown)

⇒ Pale-yellow ppt on Cold

3. Sodium Nitroprusside Test

$Na_2[Fe(CN)_5NO]$

⇒ Acetone

→ 1 ml acetone
→ 1 ml dil. NaOH
→ 1 ml Sodium Nitroprusside

⇒ Red Color → Blood

$N=N$ azo group

NH_2

Aromatic Primary amine

Basic immiscible Aniline ⇒ Amines

$\ddot{N}H_2$

* Diazotisation and Coupling

→ Primary amines $-NH_2$

$-NH-$

$-N-$

drops is

cold

1 ml aniline

1 ml HCl or H_2SO_4

1 2 3

ice

2 ml $NaNO_2$

0.25 gm α or β-naphthol + 2 ml Conc. NaOH

⇒ after 10 min ice bath

⇒ after 5 min ice bath

Reddish Brown PPT (α-naphthol)

Crimson Red PPT (β-naphthol)

Acetone

Ketones

$-\overset{O}{\overset{\|}{C}}-$ Carbonyl group

$-\overset{O}{\overset{\|}{C}}-CH_3$

sical Properties:→

re

ibility

ect on litmus paper

emical Properties:→

$-\overset{O}{\overset{\|}{C}}-$

Schiff's Reagent (shaking) → Purple Color

→ 1 ml acetone

→ 1 ml schiff's reagent

2 Iodoform Test:→

→ 1 ml acetone

→ drops of Conc. NaOH (Pale-yellow Color)

→ 3 ml I_2 (Brown)

⇒ Pale yellow ppt on Cold

3 Sodium Nitroprusside Test $Na_2[Fe(CN)_5NO]$

→ Acetone

→ 1 ml acetone

→ 1 ml dil. NaOH

→ 1 ml Sodium Nitroprusside

⇒ Red Color → Blood

N=N azo group

NH_2

aromatic Primary amine

Basic immiscible Aniline → Amines

NH_2

Primary amine: $-NH_2$

$-NH-$

$-N-$

* Diazotisation and Coupling

drop wise

dark

1 ml aniline

1 ml HCl or H_2SO_4

2 ml $NaNO_2$

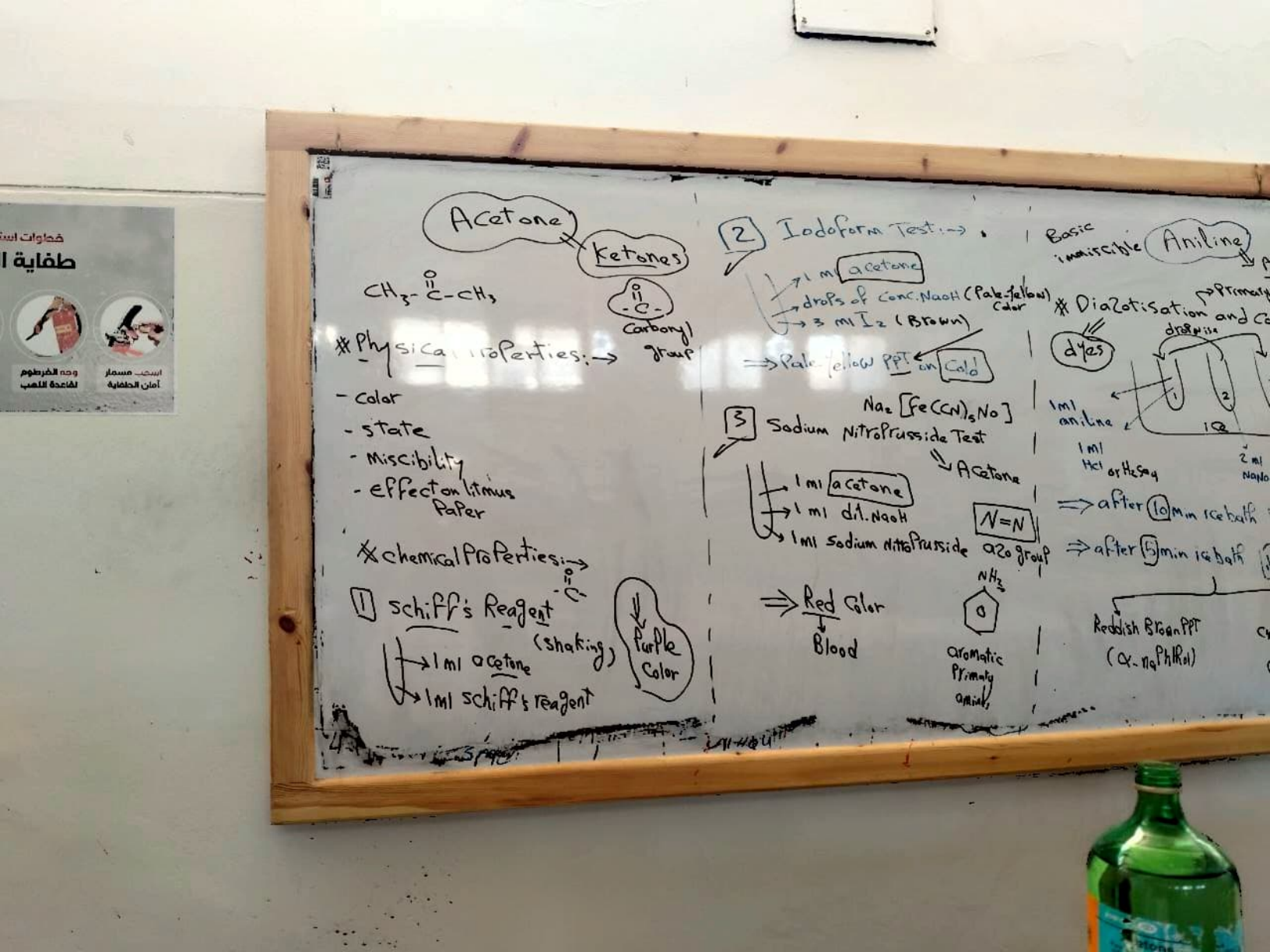
0.25 gm α or β-naphthol + 2 ml Conc. NaOH

⇒ after 10 min ice bath

⇒ after 5 min ice bath

Reddish Brown ppt (α-naphthol)

Crimson Red ppt (β-naphthol)


Acetone
Ketones
$CH_3-\overset{O}{\overset{\|}{C}}-CH_3$
-C-
Carbonyl group
* Physical Properties:→
- Color
- state
- Miscibility
- effect on litmus Paper
* chemical Properties:→
1 schiff's Reagent (shaking)
1 ml acetone
1 ml schiff's reagent
Purple Color
2 Iodoform Test:→
1 ml acetone
drops of conc. NaOH (Pale-yellow Color)
3 ml I_2 (Brown)
⇒ Pale-yellow PPT on Cold
3 Sodium Nitroprusside Test
$Na_2[Fe(CN)_5NO]$
Acetone
1 ml acetone
1 ml dil. NaOH
1 ml Sodium Nitroprusside
N=N
azo group
⇒ Red Color
Blood
NH_2
Aromatic Primary amine
Basic
immiscible
Aniline
* Diazotisation and Co
Primary
dyes
drops
1 ml aniline
1 ml HCl or H_2SO_4
2 ml NaNO
⇒ after 10 min ice bath
⇒ after 5 min ice bath
Reddish Brown PPT
(α-naphthol)

tone

Ketones

-C- (Carbonyl group)

erties:→

ties:→

-C-

ent (shaking)

e

s reagent

Purple Color

2 Iodoform Test:→

1 ml acetone

drops of Conc. NaOH (Pale-yellow color)

3 ml I_2 (Brown)

⇒ Pale yellow ppt on Cold

3 Sodium Nitroprusside Test

$Na_2[Fe(CN)_5NO]$

Acetone

1 ml acetone

1 ml dil. NaOH

1 ml Sodium Nitroprusside

⇒ Red Color

Blood

Basic immiscible Aniline

NH_2

Amines

Primary amines - NH_2

* Diazotisation and Coupling -NH- -N-

dye

drop wise

1 ml aniline

1 ml HCl or H_2SO_4

2 ml $NaNO_2$

0.25 gm α or β-naphthol + 2 ml Conc. NaOH

⇒ after 10 min ice bath

⇒ after 15 min ice bath

Reddish Brown PPT (α-naphthol)

Crimson Red PPT (β-naphthol)

N=N

azo group

NH_2

aromatic Primary amine